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Supplementary information

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The pentose phosphate pathway controls oxidative protein folding and prevents ferroptosis in chondrocytes

In the format provided by the authors and unedited

Supplementary Table 1. Top 20 marker genes in wild-type chondrocytes as shown in the heatmap in Figure 1C. RCH = resting chondrocytes; CCH = columnar chondrocytes, HCH = hypertrophic chondrocytes; PC = proliferating cells; BC = borderline cells; CC = contaminant cells.

	RCH	CCH	HCH	PC	BC	CC
1	*Meg3*	*C1qtnf3*	*Ppa1*	*Hmgb2*	*Rcan1*	*Tmsb4x*
2	*Sfrp5*	*Ndufa4l2*	*Sgms2*	*Tubb5*	*Atf3*	*Col1a1*
3	*Penk*	*H19*	*Smpd3*	*Stmn1*	*Mgp*	*Tagln2*
4	*Dcn*	*Aldoa*	*Pth1r*	*H2afz*	*Tnfrsf12a*	*Igfbp4*
5	*Dlk1*	*Fmod*	*Comp*	*Tuba1b*	*Tgfbi*	*Lgals1*
6	*Omd*	*Wwp2*	*Panx3*	*Top2a*	*Nr4a2*	*Ifitm3*
7	*Tnni2*	*Rpl32*	*Ihh*	*Cdk1*	*Klf4*	*Ier3*
8	*Itm2a*	*Rpl17*	*Cdkn1c*	*Ube2c*	*Tsc22d1*	*Mest*
9	*Frzb*	*Scrg1*	*Ccdc80*	*Cks2*	*Errfi1*	*Col4a1*
10	*Srpx2*	*Phyh*	*Col10a1*	*Birc5*	*Neat1*	*Gng11*
11	*Cst3*	*Eno1*	*Hapln1*	*Mki67*	*Cthrc1*	*S100a6*
12	*Cpe*	*Mif*	*Lgals3*	*Ccdc34*	*Nr4a1*	*Col4a2*
13	*Fibin*	*Fxyd6*	*Mef2c*	*Prelp*	*Ifrd1*	*Pfn1*
14	*Matn1*	*S100b*	*Serinc5*	*Smc4*	*Gadd45b*	*Col3a1*
15	*Efemp1*	*Rpl28*	*Slc8a3*	*Cdca3*	*Ucma*	*Junb*
16	*Zbtb20*	*Rpl35a*	*Il17b*	*Cdca8*	*Fam180a*	*Tm4sf1*
17	*Sparc*	*Lum*	*P3h2*	*Dut*	*Gadd45g*	*Nfkbia*
18	*Tubb2a*	*Aldoc*	*Fkbp11*	*Cenpa*	*Pmepa1*	*Sox4*
19	*Rbp4*	*Smoc1*	*3110079O15Rik*	*Cenpf*	*Phlda1*	*Ndufa4*
20	*Nrep*	*Rps10*	*Grb14*	*Hist1h2ap*	*Fn1*	*Crip2*

Supplementary Table 2. List of qRT-PCR primers

Gene	Primer/probe	Sequence (5' -> 3')
Acsl4	Forward primer	CCTCAGACACACCGATTCAT
	Reverse primer	CCAACTCTTCCAGTAGTGTAGTC
Cbfa1	Forward primer	TACCAGCCACCGAGACCAA
	Probe	CTTGTGCCCTCTGTTGTAAATACTGCTTGCA
	Reverse primer	AGAGGCTGTTTGACGCCATAG
Chac1	Forward primer	GCAGCGACAAGATGCCT
	Reverse primer	CACGGCTTCCCTCACATT
Col2a1	Forward primer	AGAACATCACCTACCACTGTAAGAACA
	Probe	CCTTGCTCATCCAGGGCTCCAATG
	Reverse primer	TGACGGTCTTGCCCCACTT
Col10a1	Forward primer	CCTTTCTGCTGCTAATGTTCTTGA
	Probe	TAGCCCCAAGACACAATACTTCATCCCATACG
	Reverse primer	ATGCCTTGTTCTCCTCTTACTGGA
Ctsd	Forward primer	TCACTCGAAAGGCCTACTGG
	Reverse primer	TCCACAATAGCCTCACAGCC
Ddit3	Forward primer	CACTACTCTTGACCCTGCGT
	Reverse primer	TTCCTCCTCTTCCTCCTGGG
Edem1	Forward primer	CCAGAAGGCAGTCAAGTTAGT
	Reverse primer	GGCTTCCCAGAACCCTTATC
Edem2	Forward primer	CCTGGACACCTTGCTGATTT
	Reverse primer	CTACCACTCGGATGTTGGTTTC
Ero1a	Forward primer	TGGAGGATATGGAGTGTCATCTA
	Reverse primer	GGCCTTCTAGCCAGTTGTAAA
G6pdh	Forward primer	GAAGCAGTCACCAAGAACATTC
	Reverse primer	TGGTTCGACAGTTGATTGGAG
Gpx4	Forward primer	CGCAGCCGTTCTTATCAATG
	Reverse primer	CACTGTGGAAATGGATGAAAGTC
Gpx7	Forward primer	ACCAACAGGGAGATTGAGAAC
	Reverse primer	CAGAAGTCTGGGTTAGGTACTTG
Gpx8	Forward primer	CCAACAGCTTCTACTCCTTTGA
	Reverse primer	GCAGTCACTAGCCACGTTTA
Gsr	Commercial mix	Mm.PT.56a.41678474 (IDT Technologies)

Gss	Commercial mix	Mm.PT.56a.23414710 (IDT Technologies)
Herpud1	Forward primer	CTCTGGCTACACAACGTATGG
	Reverse primer	CAGCAGTGGCAGCTAAGTAT
Hexa	Forward primer	TCGCCATTACCTGCCATTGT
	Reverse primer	AATGACCTCCTTCACATCCTGT
Hprt	Forward primer	TTATCAGACTGAAGAGCTACTGTAATGATC
	Probe	TGAGAGATCATCTCCACCAATAACTTTTATGTCCC
	Reverse primer	TTACCAGTGTCAATTATATCTTCAACAATC
Hspa5	Forward primer	GGAAAGAAGGTTACCCATGCAG
	Reverse primer	ACCCAGGTCAAACACAAGGA
Lamp1	Forward primer	TGACTCTTCCTGATGCCCTAGT
	Reverse primer	ACCTTGAAAGCCTGGACCTG
Lamp2	Forward primer	GATGGAAGCAGTTGTGGGGA
	Reverse primer	TTCACAGCCCAAGAGACAGC
Lpcat	Forward primer	TAGGACCTGCCTCATTACCT
	Reverse primer	CAGAGTGAGCCACAGGATTT
Man1a	Forward primer	CCTTGGGCATTGCTGAATATG
	Reverse primer	GTCTCCTGATAAGTGGGACAAG
Man1a2	Forward primer	GAGAAACTCCTTCCTGCCTTTA
	Reverse primer	CAGTTTCGACCTACTCCACTTT
Mmp13	Forward primer	ACTTCTACCCATTTGATGGACCTT
	Probe	CACACGCTTTTCCTCCTGGACCAAACT
	Reverse primer	AAGCTCATGGGCAGCAACA
P4hb	Forward primer	CGGTTTCTTCAAGGACGTAGAG
	Reverse primer	ACCACTGTTGGACGTGATTC
Pdia3	Forward primer	CGATGATAATGGAGAGGGGATCA
	Reverse primer	CGGTGAGTAAGTCCTTGCCT
Pdia5	Forward primer	TTAAGCTCCTGTCCACCGTG
	Reverse primer	CTTCAACGTCACAGCTCGGT
Pdia6	Forward primer	ATTGTAGATGCCGCCCTCAG
	Reverse primer	ATCACCTCTGCCCTGCTTTC
Prdx4	Forward primer	CTCCAAGCCAGCACCTTATT
	Reverse primer	GGTGAAATCCAGTGGGTAGAAG
Ptgs2	Forward primer	GGCCATGGAGTGGACTTAAA
	Reverse primer	GATACACCTCTCCACCAATGAC

Scarb2	Forward primer	TCTTCCCGTCAGACTTGTGC
	Reverse primer	TACCGAAAAGCAGGCAGTCC
Slc1a4	Forward primer	CAGTGGACTCTTTCCTCGATTT
	Reverse primer	ATGGGTCACCACTGTGTAATC
Slc1a5	Forward primer	CTGGATTCCTTTCTAGATCTCGTG
	Reverse primer	TGGATACAGGATTGCGGTATTT
Slc6a9	Forward primer	GGGAGGAGCCTTCATGTTC
	Reverse primer	AGGAAAGCTCCATGAAGAAGAG
Slc7a11	Forward primer	TTTTGTACGAGTCTGGGTGG
	Reverse primer	CGCAAGTTCAGGGATTTCAC
Slc11a2	Forward primer	GTACCTGCATTCTGCCTTAGT
	Reverse primer	ATGCAGGACTCGATGAAGAAG
Steap3	Forward primer	CTGGTCAATCTGGCTGTGAA
	Reverse primer	GGGACAGGTATATCTCCATCCT
Tfrc	Forward primer	CAGACCTTGCACTCTTTGGA
	Reverse primer	ATGACTGAGATGGCGGAAAC
Ufd1	Forward primer	GCCCAGACTTCCTGGATATTAC
	Reverse primer	CCGCAGCTCATAGATCTTCTC
Vcp	Forward primer	CATCAGCACTTCGGGAAACT
	Reverse primer	CCACAGGATACTGAACCAACTC
Xbp1-s	Forward primer	GAGTCCGCAGCAGGTG
	Reverse primer	GTGTCAGAGTCCATGGGA

Supplementary Table 3. Three-way ANOVA statistical analysis. Norm = normoxia, Hyp = hypoxia, prolif = proliferation, diff = differentiation. *p<0.05; **p<0.01; ***p<0.001. ns = non-significant.

Figure	Source of variation	Tukey's multiple comparison	Significance
S4B	G6PDH	-	***
	Diff	-	***
	Hyp	-	***
	G6PDH x Diff	-	**
	G6PDH x Hyp	-	***
	Diff x Hyp	-	***
	G6PDH x Diff x Hyp	-	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S4C	G6PDH	-	***
	Diff	-	ns
	Hyp	-	***
	G6PDH x Diff	-	**
	G6PDH x Hyp	-	***
	Diff x Hyp	-	***
	G6PDH x Diff x Hyp	-	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	*

	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
S4D	G6PDH	-	***
	Diff	-	ns
	Hyp	-	***
	G6PDH x Diff	-	ns
	G6PDH x Hyp	-	***
	Diff x Hyp	-	*
	G6PDH x Diff x Hyp	-	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***

	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S4E	G6PDH	-	ns
	Diff	-	**
	Hyp	-	***
	G6PDH x Diff	-	ns
	G6PDH x Hyp	-	***
	Diff x Hyp	-	ns
	G6PDH x Diff x Hyp	-	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***

	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S4F	G6PDH	-	ns
	Diff	-	ns
	Hyp	-	***
	G6PDH x Diff	-	*
	G6PDH x Hyp	-	*
	Diff x Hyp	-	***
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	*
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*

	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
S4G	G6PDH	-	***
	Diff	-	ns
	Hyp	-	***
	G6PDH x Diff	-	***
	G6PDH x Hyp	-	***
	Diff x Hyp	-	**
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	*
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S4H	G6PDH	-	***
	Diff	-	**
	Hyp	-	ns
	G6PDH x Diff	-	***
	G6PDH x Hyp	-	**
	Diff x Hyp	-	ns
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	**

	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S4J	G6PDH	-	ns
	Diff	-	ns
	Hyp	-	***
	G6PDH x Diff	-	**
	G6PDH x Hyp	-	*
	Diff x Hyp	-	***
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**

	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
S4O	G6PDH	-	***
	Diff	-	***
	Hyp	-	***
	G6PDH x Diff	-	***
	G6PDH x Hyp	-	*
	Diff x Hyp	-	**
	G6PDH x Diff x Hyp	-	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
S6E	G6PDH	-	***
	Diff	-	***
	Hyp	-	***
	G6PDH x Diff	-	***

	G6PDH x Hyp	-	*
	Diff x Hyp	-	ns
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	*
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
S6L	G6PDH	-	ns
	Diff	-	***

	Hyp	-	*
	G6PDH x Diff	-	ns
	G6PDH x Hyp	-	***
	Diff x Hyp	-	*
	G6PDH x Diff x Hyp	-	ns
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	*
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	**
S7E	G6PDH	-	***
	Diff	-	*
	Hyp	-	***
	G6PDH x Diff	-	***
	G6PDH x Hyp	-	ns
	Diff x Hyp	-	***
	G6PDH x Diff x Hyp	-	**
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***

	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Norm - *G6pdh*$^{chon+}$- diff	*
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Norm - *G6pdh*$^{chon-}$ - diff	**
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon+}$- diff	***
	-	Norm - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - prolif	***
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon+}$- diff	**
	-	Norm - *G6pdh*$^{chon-}$ - diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon-}$ - prolif vs. Hyp - *G6pdh*$^{chon-}$ - diff	***
	-	Hyp - *G6pdh*$^{chon+}$- diff vs. Hyp - *G6pdh*$^{chon-}$ - diff	***